

4. "With water, I say, 'Touch me not': To the tongue, I am tasteful; within limits, I am dutiful; In excess I, am dangerous!" Who am I?
 A. Carbohydrate
 B. Lipid
 C. Protein
 D. Enzyme
5. Melting point for Trans Fatty Acids is _____ in nature.
 A. Higher
 B. Lower
 C. No Change
 D. Reactive
6. Uric acid is a purine, and the end product of purine metabolism, that has been implicated in the disorder _____.
 A. Gout
 B. Hypertension
 C. Diabetes
 D. Stroke
7. If two monosaccharides differ in configuration around a single carbon atom, they are known as _____.
 A. Epimers
 B. Monomers
 C. Topomers
 D. Anomers
8. Triacylglycerol (fats) primarily stored in the _____.
 A. Liver
 B. Lung & Heart
 C. Adipose Tissue
 D. Renal
9. Nucleotide is composed of _____.
 A. Nitrogenous base + Sugar + Phosphate
 B. Nitrogenous base + Phosphate - Sugar
 C. Protein + Prosthetic Group
 D. Cofactor + Enzyme
10. The functional unit of the enzyme is known as _____.which is often made up of _____ (the protein part) and a _____ (non-protein organic part).
 A. Apoenzyme
 B. Cenzyme
 C. Substrate
 D. Holoenzyme

Identify the sentence as True or False:

(5 × 1 = 5)

(Please tick mark the answer)

True False

- | | | |
|---|--------------------------|--------------------------|
| 11. Eukaryotes possess a well-defined nucleus and are more complex in their Structure and Function. | ☐ | ☐ |
| 12. Inulin, a polymer of fructose, is used to assess renal function by measuring Glomerular Filtration Rate (GFR). | ☐ | ☐ |
| 13. Glycoproteins are a group of biochemically important compounds with a variable composition of carbohydrate (1-90%), covalently bound to protein. | ☐ | ☐ |
| 14. Major components of body molecules (obtained through intake of water fat, carbohydrates and proteins) are C, H, O, N, S. | ☐ | ☐ |
| 15. The complex carbohydrates that are not digested by the human enzymes are collectively referred to as dietary fiber. These include cellulose, hemi – cellulose, pectin, lignin, gums and mucilage. | ☐ | ☐ |

Match the following**(5 × 1 = 5)**

16.	Maltose is made up of	A. Enzymes
17.	Biocatalysts synthesized by living cells and protein in nature are known as	B. Glucose & Sucrose
18.	Sucrose	C. Amino Acids
19.	Rickets is result of deficiency of	D. Vitamin D, Calcium and Phosphorus
20.	S GPT & SGOT	E. Glucose & Glucose
		F. Mg, Vitamin A & Sodium
		G. Glucose & Fructose
		H. Alanine and Aspartate Transaminases
		I. Ozone Formations
		J. Polysaccharide

(16 -), (17-), (18-), (19-), (20-)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of B.P.T. Examination April 2018

BC102: BIOCHEMISTRY

Date: 20/04/2018

Time: 10.00 am to 1.00 pm

Maximum Marks: 50

Instructions:

1. There are two Sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
2. **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **SECTION – II** is 2 Hr 30 min.
5. Draw the figure where necessary.

SECTION-II

(Marks: 50)

Q. 1. Short answers (5 out of 7) (Answer not exceeding 50 words)

(5 × 2 = 10)

1. Define Aerobic and Anaerobic Conditions.
2. Define Ribose and Deoxyribose.
3. List out the various Liver Function Tests.
4. Omega Fatty Acids.
5. List out 20 Essential Amino Acids.
6. Define with examples: Decarboxylation & Deamination.
7. Electron Transport Chain (ETC).

Q. 2. Short Notes (4 out of 5) (Answer not exceeding 150 words)

(4 × 5 = 20)

1. Water Soluble Vitamins (Define, Specify it's Source, Deficiencies etc.).
2. Discuss the structure and functions of Disaccharides.
3. Steps of the Fatty Acid Oxidation.
4. Define Protein. Classify based on Protein Structure with example.
5. Explain the terms: Epimerization and Mutarotation (with examples)

Q. 3. Essay (2 out of 3)

(2 × 10 = 20)

1. Define and Discuss Enzyme Inhibition.
2. Describe briefly the Anaerobic Catabolism of Glucose which serves as the foundation for (TCA Cycle).
3. "I am the unit of biological activity; organized into subcellular organelles; Assigned to each are specific duties; Thus, I truly represent Life!" Identify me. Draw an outline & explain.
